DEFENSE DATA ORCHESTRATION FRAMEWORK

PLAYBOOK

Version 2.2

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CUI // FEDCON

ODT-1 | HQDA CIO/G-6 | SAIS-ADD

T2COM C2DAO

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EXECUTIVE SUMMARY

Purpose

This playbook provides comprehensive guidance for implementing the Defense Data Orchestration Framework (DDOF) across all echelons of the Department of Defense. DDOF transforms raw data into decision-ready products through a structured, repeatable six-phase lifecycle.

The Problem We Solve

Commanders waste hours hunting for data across disconnected systems. When they find it, they cannot trust it. Duplicates multiply. Security gaps proliferate. Manual approvals delay operations. DDOF fixes this.

Challenge	DDOF Solution
Hours hunting for data	Minutes to first access
30% duplicate products	Single source of truth
Unknown data quality	VAULTIS-A certified
Manual access approvals	Automated ABAC
No audit trail	Complete lineage

GENESIS MISSION ALIGNMENT

The Genesis Mission establishes data superiority as a national security imperative. DDOF directly supports three key directives:

Directive 1: Decision Dominance

American forces will possess information advantage at every echelon. DDOF compresses the OODA loop by delivering validated data products that enable faster, better decisions.

Directive 2: Bureaucracy Elimination

Data access shall not be impeded by administrative friction. DDOF automates governance enforcement and pre-authorizes access based on mission role, not individual requests.

Directive 3: Accountability

Those who steward American data will be held accountable. DDOF assigns explicit ownership at every phase with immutable audit trails.

Secretary of the Army Priorities

Priority	DDOF Implementation
Warfighter First	Every product requires operational sponsor with mission need
Speed Over Perfection	MVP delivery within 30 days; iterate based on feedback
Cut the Fat	Mandatory duplicate check before new development
Trust the Troops	Phase 1 approval delegated to battalion-level C2DAO

FRAMEWORK OVERVIEW

The Six Phases of DDOF

DDOF consists of six sequential phases. Each phase has defined inputs, activities, outputs, and quality gates. No phase can begin until the previous gate is passed.

Phase	Name	Duration	Key Output
1	Problem Framing	1-5 days	Approved requirement
2	Data Provisioning	3-10 days	Secured data access
3	Data Wrangling	5-15 days	Clean, integrated data
4	Development	10-30 days	Functional product
5	Test & Evaluation	5-10 days	Validated product
6	Operations	Ongoing	Live, maintained product

What DDOF Is

- A process framework adaptable to any data problem
- Platform-agnostic (works with Vantage, ADVANA, or any approved system)
- Scalable from battalion dashboards to enterprise analytics
- Iterative with continuous improvement feedback loops
- Governed with explicit quality gates at each phase

What DDOF Is NOT

- Not a specific tool or platform
- Not a one-time certification
- Not optional for enterprise data products
- Not a replacement for governance—it operationalizes it

ROLES AND RESPONSIBILITIES

Role	Echelon	Primary Function
Decision Maker (DM)	O-6+	Articulates requirements, approves final products
C2DAO	O-4/O-5	Governance oversight, quality assurance, gate approvals
Functional Data Manager (FDM)	O-3/O-4	Product ownership, lifecycle management
Data Engineer	Technical	Build and maintain data pipelines
Data Scientist/ORSA	Technical	Analytical modeling, algorithm development
Knowledge Manager	Staff	Documentation, training, organizational learning

PHASE 1: PROBLEM FRAMING

Purpose

Phase 1 transforms an operational need into a clearly defined, approved data requirement. This phase prevents wasted effort by ensuring alignment between decision-maker needs and technical capabilities before development begins.

Key Steps

- **Step 1.1:** Requirement Validation — Search ADC for existing products, validate operational context
- **Step 1.2:** Problem Statement Development — Create SMART problem statement
- **Step 1.3:** Analytical Approach Selection — Determine technical methodology
- **Step 1.4:** C2DAO Review and Approval — Obtain formal gate approval

SMART Criteria

Criterion	Definition	Example
Specific	What exactly do you need?	Equipment readiness by unit
Measurable	How will you know it's accurate?	Match GCSS-A within 2%
Achievable	Can it be built?	Platform and data exist
Relevant	Does it support a decision?	Tied to METL task
Time-bound	When must it be ready?	IOC by 15 JAN

Gate 1 Exit Criteria

- Requirement Validation Memo signed
- Problem Statement approved
- Analytical Approach documented
- C2DAO Gate 1 approval received
- ADC intake record created

PHASE 2: DATA PROVISIONING

Phase 2 secures access to authoritative data sources. This phase ensures data is legally accessible, properly classified, and technically retrievable.

Key Steps

- Data Requirements Specification
- Source Authority Verification
- Access Request and Security Coordination
- Pipeline Establishment

Gate 2 Exit Criteria

- All sources verified as authoritative
- Access secured to all required sources
- Provisioning pipeline operational
- Data flowing to development environment

PHASE 3: DATA WRANGLING

Phase 3 transforms raw data into analysis-ready form through cleaning, validation, transformation, and integration. Data must meet VAULTIS-A standards.

VAULTIS-A Quality Framework

Dimension	Definition	Standard
Visible	Clearly marked and discoverable	100% visibility in catalog/product
Accessible	Usable by authorized personnel	99% access rate for authorized users
Understandable	Clearly documented and interpretable	Complete metadata and user guide
Linked	Relationships maintained	100% linkage to sources/products
Trusted	Provenance and quality validated	95%+ accuracy, sponsor sign-off
Interoperable	Usable across platforms/systems	90%+ compatibility with approved platforms
Secure	Protected per classification	100% compliance with security policy
Auditable	Complete lineage available	Full provenance and access logs

Gate 3 Exit Criteria

- EDA complete and documented
- Cleaning rules defined and tested
- Quality validation passed (≥85%)
- Lineage documented

PHASE 4: DEVELOPMENT

Phase 4 builds the data product. This phase transforms integrated data into operational tools: dashboards, models, reports, or APIs.

MVP Timeline Mandate

Per Secretary of the Army priorities, MVP delivery within 30 days is the standard. Extensions require C2DAO approval with documented justification.

Required Governance Controls

- Access control (ABAC/RBAC per classification)
- Audit logging (all queries and changes)
- Data lineage (automated capture)
- Classification marking (visible in product)
- Ownership display (FDM contact visible)

Gate 4 Exit Criteria

- Product functional
- Governance controls implemented
- User Guide drafted
- Development QA passed

PHASE 5: TEST & EVALUATION

Phase 5 validates that the product meets requirements, is fit for operational use, and provides accurate, reliable information.

Test Types

- **Functional:** Verify features work as specified
- **Data Quality:** Verify accuracy, currency, completeness
- **Security:** Verify access controls before deployment
- **UAT:** Obtain end-user validation

Gate 5 Exit Criteria

- Functional testing passed
- Data quality testing passed (≥85%)
- UAT completed with sponsor sign-off
- Security validation passed
- All critical defects resolved

PHASE 6: OPERATIONS

Phase 6 deploys the product to production, registers it in the Army Data Catalog, and establishes ongoing operations.

Key Activities

- Production deployment
- ADC registration with complete metadata
- Operations handoff (runbook, escalation)
- User training
- Continuous improvement feedback loops

ADC Registration Requirements

- Product name and description
- Owner (FDM) contact information
- Classification and handling
- Data sources and lineage
- Refresh frequency
- Access method
- VAULTIS-A quality score

Gate 6 Exit Criteria

- ■ Product deployed to production
- ■ ADC registration complete
- ■ Operations handoff complete
- ■ Training delivered
- ■ Monitoring operational

GOVERNANCE ENFORCEMENT

Quality Gates Are Mandatory

Gates prevent progression without verified compliance. They are enforced, not advisory. No exceptions without C2DAO override and audit trail.

Fail-Closed Enforcement

Per Genesis Mission directives, data products must fail closed—denying access when authorization cannot be confirmed. Default deny. Explicit authorization required. All denials logged and audited.

Duplicate Prevention

ADC search is mandatory before any new development. Duplicate products waste resources and create confusion. If similar product exists, modify it rather than creating new.

Data Product Retirement

- No access in 90 days → FDM review required
- No access in 180 days → Retirement initiated
- Source data discontinued → Retirement initiated
- Quality score drops <70% → Remediate or retire

QUICK REFERENCE GUIDE

Phase Timeline Summary

Phase	Standard	Maximum
Phase 1: Problem Framing	3-5 days	10 days
Phase 2: Data Provisioning	5-10 days	30 days
Phase 3: Data Wrangling	7-12 days	20 days
Phase 4: Development	15-25 days	35 days
Phase 5: Test & Evaluation	7-12 days	20 days
Phase 6: Operations	Ongoing	Ongoing
TOTAL (IOC)	**30-60 days**	**90 days**

Key Contacts

- C2DAO: Your command's designated Data Analytics Officer
- ADC Support: Army Data Catalog help desk
- Platform Support: Vantage/ADVANA/Gabriel Nimbus teams
- Policy Questions: ODT-1, HQDA CIO/G-6

Configuration Management Friction Matrix

	Wedge 1: Source Systems -> Platform Integration		Wedge 2: Platform Integration		Wedge 3: Platform Integration -> Data Products	
	Problem Framing	Data Provisioning	Data Wrangling	Development	Test & Evaluation	Operations
Overall Phase Status	Status: OK/ FRICTION / CASCADING FRICTION	Status: OK/ FRICTION / CASCADING FRICTION	Status: OK/ FRICTION / CASCADING FRICTION	Status: OK/ FRICTION / CASCADING FRICTION	Status: OK/ FRICTION / CASCADING FRICTION	Status: OK/ FRICTION / CASCADING FRICTION
People	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…
Process / Policy	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…
Products	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…
Resources	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…
Technology	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…	Status: OK/ FRICTION / CASCADING FRICTION Describe friction here…

GLOSSARY

Term	Definition
ABAC	Attribute-Based Access Control
ADC	Army Data Catalog
ADS	Authoritative Data Source
C2DAO	Command Data Analytics Officer
CDAO	Chief Data and Analytics Officer
CUI	Controlled Unclassified Information
DDOF	Defense Data Orchestration Framework
ETL	Extract, Transform, Load
FDM	Functional Data Manager
FOC	Full Operational Capability
IOC	Initial Operational Capability
MVP	Minimum Viable Product
ORSA	Operations Research Systems Analyst
UAT	User Acceptance Testing
VAULTIS-A	Visible, Accessible, Understandable, Linked, Trusted, Interoperable, Secure, Auditable

— END OF PLAYBOOK —

"Data is ammunition. The DDOF ensures every round is accurate, available, and authorized."